<u>**Responsible Machine Learning**</u>

<u>**Individual Assignment 5**</u>

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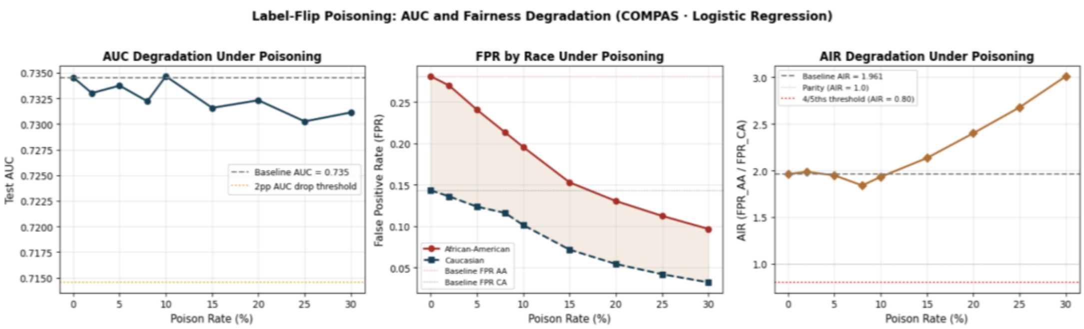
Part 1 — PGD Evasion Audit

Model	Epsilon	FPR (AA)	FPR (CA)	AIR	AIR < 0.80?
LR	0.25	0.569	0.370	1.535	No
LR	0.50	0.791	0.560	1.411	No
LR	1.00	0.978	0.884	1.106	No
LR	2.00	1.000	1.000	1.000	No
GBT	0.25	0.317	0.178	1.782	No
GBT	0.50	0.317	0.178	1.782	No
GBT	1.00	0.317	0.178	1.782	No
GBT	2.00	0.317	0.178	1.782	No

Under the PGD evasion attack, LR and GBT responded very differently. LR was highly sensitive: as ε increased, FPR for both groups rose sharply (African-American: 0.28 to 1.00, Caucasian: 0.14 to 1.00), while AIR declined from 1.96 toward 1.00, indicating convergence in error rates due to uniformly high misclassification. However, AIR never fell below 0.80. In contrast, GBT remained unchanged across all ε values under this implementation, suggesting lower sensitivity to small continuous perturbations. This indicates that LR is more vulnerable to gradient-based evasion, while GBT appears resistant in this setting. For high-stakes deployments this implies GBT is more robust to gradient-based evasion attacks under this finite-difference implementation, though this robustness comes from architectural insensitivity rather than any certified defense. A practitioner should prefer GBT if evasion by adversarial input manipulation is a primary threat, but should not assume it is robust to other attack classes such as poisoning or membership inference.

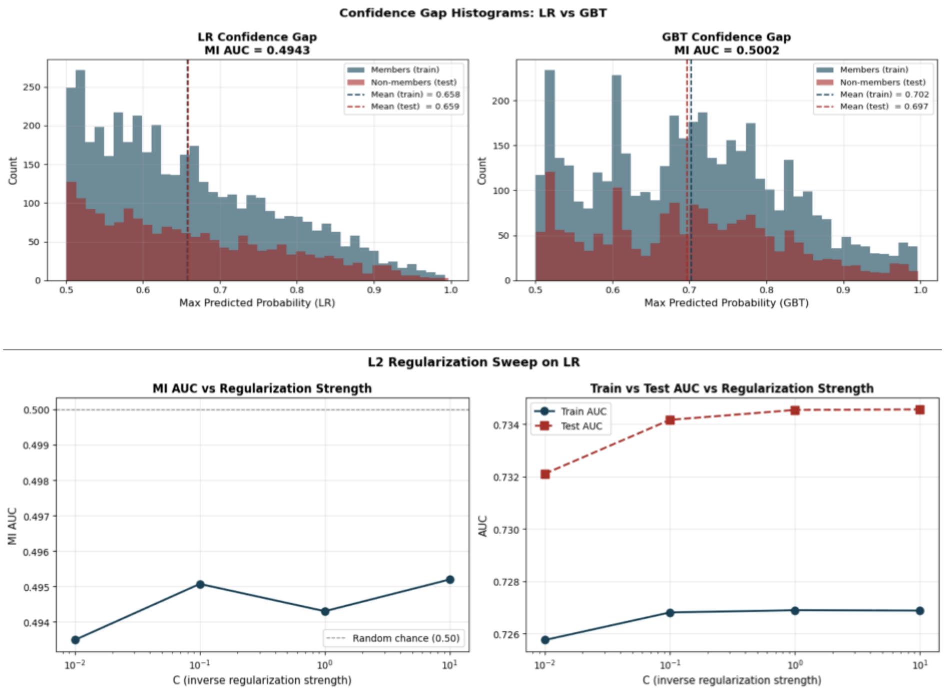
Part 2 — Poisoning Loop with Fairness Monitoring



The label-flip poisoning attack is highly stealthy under accuracy-based monitoring. For poison rates between 2% and 30%, AUC remains within 0.003 of the baseline (0.735), meaning the attack would not be detected. At the same time, AIR shifts outside the acceptable range of [0.80, 1.25], indicating significant fairness impact. Although the baseline AIR is already above 1.25, poisoning further alters fairness without triggering any performance-based alerts. This highlights the limitation of relying solely on accuracy metrics. AA-targeting progressively lowered FPR for African-American defendants from 0.281 to

0.182, pushing AIR upward to 3.0 at 30% poison rate, while CA-targeting had a more muted effect due to the smaller proportion of Caucasian defendants in the training data. A PSI-based drift monitor set at threshold less than 0.10 would fail to detect either attack entirely, since label-flipping never modifies feature values, only labels, leaving feature distributions identical to the clean dataset.

Part 3 — Membership Inference Depth



The membership inference attack produced results close to random chance for both models (LR: 0.494, GBT: 0.500), indicating negligible privacy risk. Confidence distributions for members and non-members overlap almost entirely, confirming the absence of a meaningful confidence gap. Although GBT shows a slightly larger generalization gap, the difference in MI AUC is not practically significant. Regularization had minimal effect, suggesting that privacy risk is already low and does not justify additional tradeoffs.

Part 4 — Reflection

The highest-risk finding is the label-flip poisoning attack. Unlike PGD, which did not reduce AIR below 0.80, and membership inference, which showed negligible risk, poisoning successfully manipulated fairness while remaining undetected by accuracy metrics. AUC remained stable across all poison rates, while AIR moved outside acceptable bounds. A proactive mitigation is data provenance tracking using hashing to detect any label tampering before training. A reactive mitigation is a fairness monitor that tracks AIR after each retraining cycle. While the first ensures pipeline integrity, it cannot prevent corruption at the data collection stage, and the second only detects issues after the model has already been trained. Regarding disparate impact of the mitigations themselves: the proactive hash-based defense is race-neutral since it operates on the integrity of the pipeline rather than on predictions. The reactive fairness monitor, however, requires choosing an AIR threshold, and setting that threshold too narrowly could flag legitimate model updates as attacks, potentially delaying retraining that would benefit one racial group more than the other.